

# Optimization – 1

Extremal problems: numbers, polynomials, functionals

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## Theoretical minimum.

**Theorem 1 (existence).** A continuous function on a compact set attains its extrema. On a non-compact set nothing is guaranteed, and for a functional on a class of functions compactness is never free: a bounded sequence of functions need not have a convergent subsequence. Therefore “find the maximum” splits into two independent tasks — produce a candidate and prove that the supremum is attained.

**Theorem 2 (where a maximum can sit).** An interior extremum of a differentiable function is a critical point; on a set given by constraints the extremum may lie on the boundary, at a corner or at a point where the constraints are degenerate. Checking all three kinds of points is part of the answer, not an afterthought.

**Fact 1 (the three rungs of the ladder).** The problems below are ordered by what one optimises over: a number, then a polynomial from a class with a uniform bound, then a function from a class defined by integral constraints. On the second rung the extremal object is recognised by an alternation property, as in Chebyshev’s theorem; on the third rung the machinery is developed in Optimization – 2 and – 3, and here we solve the problems by hand.

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1. (Putnam 1986) Find the maximum value of  $f(x) = x^3 - 3x$  on the set of all real  $x$  satisfying  $x^4 + 36 \leq 13x^2$ .
2. (Putnam 2003) Find the minimum value of  $|\sin x + \cos x + \operatorname{tg} x + \operatorname{ctg} x + \sec x + \operatorname{cosec} x|$  for real  $x$ .

3. (Putnam 1968) Let  $V$  be the collection of all quadratic polynomials  $P$  with real coefficients such that  $|P(x)| \leq 1$  for all  $x \in [0, 1]$ . Determine  $\sup\{|P'(0)| : P \in V\}$ .
4. (Putnam 2016) Find the smallest constant  $C$  such that for every real polynomial  $P(x)$  of degree 3 having a root in  $[0, 1]$  one has  $\int_0^1 |P(x)| dx \leq C \max_{x \in [0, 1]} |P(x)|$ .
5. (VJIMC 2009) Let  $E$  be the set of all continuously differentiable  $f$  on  $[0, 1]$  with  $f(0) = 0$  and  $f(1) = 1$ , and let  $J(f) = \int_0^1 (1+x^2)(f'(x))^2 dx$ .
  - a) Show that  $J$  attains its minimum on  $E$ .
  - b) Find that minimum and the function realising it.